# Testing is Critical for the Adoption of Software-Defined Autonomous Vehicles

About 94 percent of serious crashes are due, in part, to frequent and predictable driver errors, such as speeding, or driving while impaired or distracted.

Developments in the autonomous vehicle (AV) arena are accelerating as traditional automakers, along with new players, invest in technology that is driving innovation. While AVs have the potential to improve automobile safety and driving convenience, their complexity requires a rigorous test and verification system to ensure vehicle safety across all types of traffic, roads, and weather conditions. Certainly, AVs will use methods based on artificial intelligence (AI), which will enable vehicles to communicate via telecom service and infrastructure providers.

Meanwhile, ADAS and AV developments are shifting to be software-defined. The Software Defined Vehicle (SDV) transformation is the shift from hardware-centric to software-driven automotive design. In SDVs, key functions such as braking, steering, infotainment, and driver assistance are increasingly controlled and updated through software. This enables over-the-air (OTA) updates, continuous feature improvements, and personalized user experiences. Automakers are adopting centralized computing architectures that separate hardware from software, enabling faster innovation and scalability. The SDV transformation is revolutionizing the auto industry by making vehicles smarter, more connected, and adaptable over their lifetime. This evolution will enable manufacturers to continually improve vehicle performance, safety, and user experience through software enhancements, without requiring major hardware changes. By leveraging data analytics and connected sensors, these vehicles can adapt to changing conditions and customer needs in real time.

Overall, the transformation enhances flexibility, reduces maintenance costs, and paves the way for innovations like autonomous vehicles.

Autonomous vehicle (AV) technology is fundamentally built on the concept of the connected car—where external systems and infrastructure communicate with the vehicle to provide real-time data on traffic, road conditions, and surrounding vehicles. AV systems integrate a suite of sensors, high-performance computing, and advanced software algorithms to enable perception, decision-making, and control. As a result, self-driving vehicles are already demonstrating statistically safer performance per mile driven compared to human-operated vehicles, underscoring their potential to improve road safety and efficiency.

According to a 2024 American Automobile Association (AAA) survey, 66 percent of U.S. drivers report being afraid to ride in fully self-driving vehicles, while an additional 25 percent express uncertainty. This level of apprehension has remained high, reflecting persistent public concern over the safety and reliability of autonomous vehicle technology. [1]

Safety concerns are top of mind based on the number of accidents that are caused by driver error. Safety improvements usually top the list of potential benefits of AVs. Eliminating human error from the driving equation significantly reduces traffic injuries and fatalities.

1. AAA Newsroom – March 14, 2024

# Overview of Automation

Today, a range of automation options are available to assist drivers, and many are already on the market. To put things into perspective, the Society of Automotive Engineers (SAE) has established levels of automation [2] for AVs as shown in Figure 1.

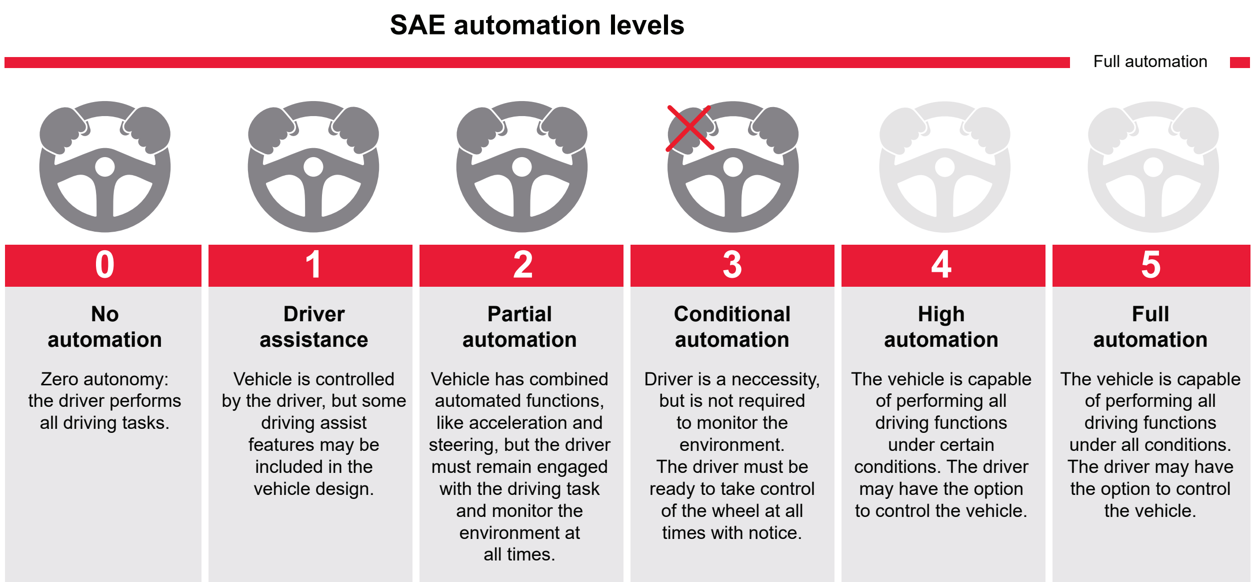



**Figure 1.** SAE Automation Levels describe the different levels of autonomous vehicle capability

- Level 0 is no automation at all; the driver does all of the work.
- Level 1 adds some driver-assistance features, such as adaptive cruise control and blindspot monitoring.
- Level 2 still requires the driver to drive but offers both steering assistance and speed control features.
- Level 3 automates the driving of the vehicle but still requires the human driver to remain attentive and take control within a specified amount of time.
- At Levels 4 and 5, the AV truly becomes autonomous. For example, Level 4 automated driving limitations are restricted to certain conditions such as specific geography or route, weather, traffic type, speed, and roadway. Level 5 represents a no restrictions self-driving car.

2. https://driverless.wonderhowto.com/news/definitive-guide-levels-automation-for-driverless-cars-0176009/

The automobile industry is rapidly adding features to vehicles at Levels 2 and 3, and there are many opportunities to assist the driver while maintaining the existing model in which the driver actively controls the car. A wide range of sensors (cameras, radar, lidar, and ultrasonic sensors) let the vehicle “see” what is going on around it and automatically assists the driver. Many vehicles already offer driver-assisting features such as blind-spot monitoring, backup cameras, automatic high beam headlights, adaptive cruise control, lane-keeping assistance, and automatic braking.

Levels 1 through 3 can improve automobile safety, but Levels 4 and 5 are required to deliver additional potential benefits including mobility improvements for disabled and senior adults, increases in personal productivity, and new transportation models. This makes Levels 4 and 5 the ultimate goal of most AV programs to leave the driving to the car.

Level 4 is a natural precursor to Level 5 and limits the operational design domain (ODD); the AV is autonomous under certain specific conditions. For example, a Level 4 AV may only handle specific types of roads — limited access highways, HOV lanes, AV-only lanes, rural roads, or closed campuses. There may be restrictions depending on visibility; there is no AV action in extreme weather. Another constraint might be specific infrastructure support for AVs or roads that are pre-mapped.

Deployments are typically local or regional. Hitting 100 percent coverage of all scenarios is difficult. It’s not too hard to get to 95 percent, but the last 5 percent is challenging; so reducing the ODD has a significant impact. For example, a Level 4 use case might be an autonomous taxi service in a confined, well-mapped area with known streets, lower speeds, and extreme weather to deliver the minimum required visibility. Another possible deployment is long-haul cargo trucks limited to specific, known routes.

Level 5 automation poses the challenge of handling every possible driving situation. Due to the scalability of software-enabled systems, Level 5 automation enables the vision of replacing the average driver with the most capable software-defined driver in the world. Due to technological, regulatory, and economic challenges, adoption is expected to be more gradual than previously thought. However, we are seeing more rapid advancements and deployments in autonomy across several categories, including robotaxis, autonomous trucks, and geofenced industries such as mining and farming.

> “Reaching L4 autonomy remains significantly constrained by both technical and regulatory barriers.”
>
> World Economic Forum May 2025 [3]

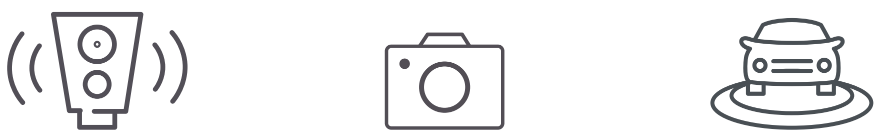
3. www.weforum.org/stories/2025/05/autonomous-vehicles-technology-future/

# Sensor and Communication Technologies

A combination of new technologies working together makes AVs possible — sensors, computing power, smart software, communications, and navigation. AVs use sensors to see the world around them, just like human drivers. Perhaps, better than humans because they can see in all directions simultaneously.

AVs use three main types of sensors to monitor the driving environment:

- Vision with cameras
- Radar detection and ranging system
- Ultrasonic



| | **Radar** | **Camera** | **Ultrasonic** |
|---|---|---|---|
| Technology | Detection – distance (range) and motion (velocity and angle) by radio waveforxms | Recognition, classification by images | High-frequency sound waves bounce off of nearby objects, and distance is calculated |
| Application | • Adaptive cruise control<br>• Automatic emergency braking systems<br>• Blind spot detection | • Traffic sign recognition<br>• Lane keep systems<br>• Parking assistance<br>• Blind spot detection<br>• ACC, AEBS<br>• Surround view<br>• Rear collision warning<br>• Cross traffic alert | Detect obstacles and improve vehicle maneuverability, parking assistance, blind spot detection, lane change assistance, navigation |

**Figure 2.** Autonomous vehicles use an array of sensors to see the driving environment.

Vision systems, powered by image sensors, require significant processing to extract useful information With cameras playing a vital role in detecting the vehicle's immediate surroundings. Radar sensors operate 76-71 GHz and are positioned in the front or rear of the vehicle to monitor traffic and obstacles. They can detect objects from a few centimeters to several hundred meters away. Radar is particularly useful in poor weather conditions, when a camera is less effective.  Ultrasonic sensors complement these two by detecting nearby objects, making them ideal for low-speed maneuvers like parking and reversing.

Wireless communications play another important role in AV functionality, enabling vehicles to exchange information with other vehicles (V2V), pedestrians (V2P), or roadside infrastructure (V2I). Often, these are available in vehicle-to-everything (V2X). These communications channels provide essential information to the vehicle, including notifications of traffic congestion and road hazards.

Since no single sensor can provide a complete understanding of the environment, AVs employ sensor fusion—combining data from cameras, radar, communications, and ultrasonic sensors—to create a comprehensive, real-time model of their surroundings. This integration significantly boosts both safety and reliability.

To support these capabilities, AVs generate and process vast amounts of data. At the heart of this system is artificial intelligence, which interprets sensor inputs and powers the vehicle's decision-making and navigation functions, enabling it to respond intelligently to dynamic driving conditions.

# Strict Testing is Critical to Gain Acceptance

All new technologies face barriers to adoption and AVs are no exception. Because of the safety issues involved, adoption of AVs will initially encounter resistance by many consumers.

Public opinion, familiarity, and trust play a significant role in the consumer's willingness to adopt AV technology. Consumer perceptions are likely to shift over time, as AV technology is proven and becomes more familiar. AV testing is critical to validate self-driving cars are safe enough to be onthe road. Adoption of Levels 2through 4 is happening incrementally as automakers add more driver assistance features to their vehicles. Level5 represents a bigger challenge because the driver is removed from the system, leaving the AV to drive on its own.

# Validating a Complex System in a Complex Environment

All systems have a non-zero failure rate, so while you would like to design an AV that never makes a wrong decision, the real question is how good does it have to be? You expect that an AV design will perform better than the average human driver. The industry is grappling with how good is good enough; and how will you know that level of reliability is achieved.

While referring to the number of miles driven may give us a sense of how much testing needs to take place, it is not a reliable metric for describing the robustness of the testing. A mile of testing on a limited-access rural highway is quite different than a mile in a complex urban environment.

Specifically, you must ensure testing of critical edge cases — challenging scenarios that only rarely occur in normal traffic but may be deadly when they happen.

Experience with complex systems for both hardware and software includes quality and reliability standards from the start. Attempting to take a weak design and test your way to quality does not produce top quality results.

An effective validation program begins with a test strategy that considers the entire system's operation. Electronic and software-based components have been a part of modern vehicles for many years, so the automotive industry has extensive experience in designing and verifying system reliability. Current test strategies employ a layered approach, with verification performed at various levels of abstraction within the system. Software-defined vehicle development will bring about change to the way testing is done. Integrating testing into every stage of the pipeline with a test methodology known as TestOps, automating testing across the modules, systems and in-vehicle integration. Continuous validation with simulation as well as hardware-in-the-loop environments.

Additionally, several regulatory standards are in place to require that the systems used are tested and are safe. ISO 26262 is an international standard that addresses the functional safety of electrical and electronic systems in automotive production. It is a risk-based safety standard, where the risk of hazardous operational situations is qualitatively assessed, and safety measures are defined to avoid or control systematic failures and to detect or control random hardware failures or mitigate their effects.

UNECE (United Nations Economic Commission for Europe) Regulations R155 and R156, introduced by the UN's World Forum for Harmonization of Vehicle Regulations (WP.29), are becoming global standards for automotive cybersecurity. Although officially required in regions such as Europe, Japan, Australia, and New Zealand, most automakers worldwide are aligning with them. These rules indirectly impact component suppliers through obligations placed on vehicle manufacturers.

R155 mandates a Cybersecurity Management System (CSMS) to protect vehicles from cyber threats, while R156 requires a Software Update Management System (SUMS) to ensure ongoing security throughout a vehicle's lifecycle. These regulations support the adoption of standards like ISO/SAE 21434 and are essential for securing connected vehicles in today's digital automotive landscape. [2]

# System Level Verification

Testing vehicles under a wide range of environmental, road, and traffic conditions is the ultimate test of AV system performance. Ideally, this would cover every conceivable driving scenario to ensure the AV can handle them. Public road testing by AV companies has received much publicity because it is visible to the general public. This type of road testing is invaluable because it exposes the vehicle to a wide range of real-world scenarios. AV companies also use private test tracks which offer a controlled and repeatable environment but with less variation in test scenarios. Virtual test tracks are a critical tool for generating a variety of repeatable test scenarios at a reasonable cost.

Simulated driving conditions play a significant role in the test strategy for AVs. Because sensors and actuators send and receive digital data, their sensed world is captured and replayed as digital streams of data. Use of real-world data to feed simulations will enhance test coverage and corner cases. A virtual representation of the AV, complete with virtual sensors and actuators, is tested in a virtual world driven by the same software as the real-world autonomous vehicle. As the vehicle operates in this environment, models replicate what the vehicle "sees" in the real world. This virtual testing methodology recreates the AV driving scenarios and can be "driven" for millions of miles at a lower cost while offering more repeatable results compared to actual road tests.

2. https://www.keysight.com/blogs/en/tech/educ/automotive-cybersecurity

# The Software-Defined Shift: Building Trust Through Testing

The shift to SDVs marks a transformation from hardware-centric to software-driven vehicle design, enabling continuous updates, improved performance, and personalized user experiences. It also supports the integration of advanced driver-assistance systems (ADAS) and paves the way for fully autonomous driving. Continuous validation throughout the development pipeline, known as TestOps, is a critical part of this transition. Vigorous testing of these complex systems is critical to verify their safety and will determine whether AVs are safe enough to be on the road.

Keysight Technologies is partnered with AV developers to support this mission by providing advanced testing solutions. The success of autonomous vehicles hinges not only on technological innovation but also on the industry's ability to demonstrate safety and reliability through comprehensive and transparent testing practices.